BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2025

PHYSICS

Past Paper 2025

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Physics (Objective Type) - Group - I

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Half life of an isotope is one day, what quantity remains after 2 days:

(A) Half
(B) One quarter
(C) One eighth 1/8
(D) None of these

Q2. Presence of magnetic field can be detected by a:

(A) Small mass
(B) Stationary positive charge
(C) Stationary negative charge
(D) Magnetic compass

Q3. Ohm's law is:

(A) $I \times V = R$
(B) $V = R/I$
(C) $V = I/R$
(D) $V = IR$

Q4. If mass of bob is increased by 3 times then time period of simple pendulum:

(A) Increased
(B) Same
(C) Decreased
(D) All of these

Q5. Refractive index of ice is:

(A) 1.30
(B) 1.31
(C) 1.32
(D) 1.33

Q6. Which form of energy is sound:

(A) Electrical (B) Mechanical

(C) Thermal (D) Chemical

Q7. One byte is equal to:

(A) 8 bits (B) 16 bits

(C) 64 bits (D) 128 bits

Q8. A positive electric charge:

(A) Attract positive (B) Attract neutral

(C) Repel positive (D) Repel neutral

Q9. It does not change during refraction of light:

(A) Speed (B) Frequency

(C) Direction (D) Wavelength

Q10. The output of NAND gate is zero if :

(A) B = 0, A = 0 (B) B = 1, A = 1

(C) A = 0, B = 0 (D) A = 1, B = 1

Q11. Equivalent resistance in series is 8Ω, what will be in parallel :

(A) 2Ω (B) 4Ω

(C) 8Ω (D) 12Ω

Q12. In computer terminology, information means:

(A) Any data (B) Raw data

(C) Processed data (D) Large data

Physics (Subjective) (PART - I) - Group - I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Differentiate between compression and rarefaction.

(ii) Write two properties of simple harmonic motion.

(iii) What is spring constant? Write its formula.

(iv) Does electric potential depend upon test charge?

(v) How would you define potential difference between two points? Define its units.

(vi) What is meant by Faraday Cage?

(vii) Write the symbol and truth table of NAND gate.

(viii) What do you know about discovery of electron?

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Write two methods to reduce noise pollution.

(ii) Define intensity level of sound, write its unit.

(iii) Differentiate between noise and music.

(iv) What is current also write its formula.

(v) Define unit of resistance.

(vi) Describe Joule's law also write its formula.

(vii) Define fax machine.

(viii) Write down two uses of internet.

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) State what is difference between regular and irregular reflection?

(ii) What is mirror formula? Write its mathematical form.

(iii) An object is placed 6 cm in front of a concave mirror that has focal length 10 cm. Determine the position of image.

(iv) Define right hand grip rule.

(v) What is D.C. motor?

(vi) State the principle of an A.C. generator.

(vii) What is the atomic number, write its symbol.

(viii) Find the number of protons and neutrons in nuclide defined by $^{13}_{6}X$.

Physics (Subjective) (PART - II) - Group - I

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5(a). (Marks 4) Describe construction of ripple tank and hence explain phenomenon of reflection and diffraction.

Q5(b). (Marks 5) The charge [illegible in source scan] many negatively charged particles would be [illegible in source scan] 100 μC. Assume charge on one negatively charged particle is 1.6×10^{-19} C.

Q6(a). (Marks 4) Explain the parallel combination of resistors with the help of diagram.

Q6(b). (Marks 5) A marine survey ship sends a sound wave straight to the seabed. It receives an echo 1.5 s later. The speed of sound in [illegible in source scan] is 1500 ms^{-1}. Find the depth of the sea at [illegible in source scan].

Q7(a). (Marks 4) What is transformer? State the principle of its working? Write a note on its types.

Q7(b). (Marks 5) A convex lens of focal length 6 cm is to be used to form a virtual image three times the size of the object. Where must the lens be placed?

GROUP – II

Physics (Objective Type) - Group - II

Time Allowed: 15 Min. | **Maximum Marks:** 12

Note: Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The relation between v, f and λ of a wave is:

(A) $vf = \lambda$ (B) $f\lambda = v$

(C) $v\lambda = f$ (D) $v = \lambda f$

Q2. Which form of energy is sound:

(A) Mechanical (B) Electrical

(C) Thermal (D) Chemical

Q3. S.I unit of capacitance of a capacitor is:

(A) V (volt) (B) A (ampere)

(C) F (farad) (D) N (newton)

Q4. Which of the following quantity is not changed during refraction of light:

(A) Its direction (B) Its speed

(C) Its wavelength (D) Its frequency

Q5. Five joules of work is needed to shift 10 C of charge from one place to another. The potential difference between the places is:

(A) 0.5 V (B) 2 V

(C) 5 V (D) 10 V

Q6. An electric current in conductors is due to the flow of:

(A) Positive ions
(B) Negative ions
(C) Positive charges
(D) Free electrons

Q7. The direction of induced e.m.f. in a circuit is in accordance with conservation of:

(A) Mass
(B) Energy
(C) Charge
(D) Momentum

Q8. Half-life of carbon-14 is:

(A) 123 years
(B) 30 years
(C) 5730 years
(D) 7530 years

Q9. The process by which electrons are emitted by a hot metal surface is known as:

(A) Boiling
(B) Evaporation
(C) Conduction
(D) Thermionic emission

Q10. What does the term e-mail stand for:

(A) Emergency mail
(B) Electronic mail
(C) Extra mail
(D) External mail

Q11. Release of energy by the sun is due to:

(A) Nuclear fission
(B) Chemical reaction
(C) Nuclear fusion
(D) Burning of gases

Q12. Formula for the electric potential is:

(A) $V = W/q$
(B) $V = qW$
(C) $V = q/W$
(D) $V = 2qW$

Physics (Subjective) (PART - I) - Group - II

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

Q2. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Find the time period of simple pendulum of 1.0 m long where $g=10ms^{-2}$.

(ii) Define transverse wave with example.

(iii) What are damped oscillations?

(iv) Define volt.

(v) What is capacitor? Write the unit of capacitance.

(vi) If V = 50 V and C = 100μF then Q = ?

(vii) Draw symbol diagram of AND gate. Write down its truth table.

(viii) Write the names of logic operations.

Q3. (Marks 10) Write short answers to any FIVE (5) questions:

(i) What is meant by reverberation?

(ii) Define pitch, on what factors it depend.

(iii) Why sound of women is shrill as compared to men?

(iv) What is unit of electric current, define it.

(v) Define e.m.f., also write its formula.

(vi) Write down the limitations of Ohm's law.

(vii) Define transmitter and transmission channel.

(viii) What do you know about flash drive?

Q4. (Marks 10) Write short answers to any FIVE (5) questions:

(i) Define Snell's law. Write down its formula.

(ii) Differentiate between convex mirror and concave mirror.

(iii) Find the value of critical angle for water (refracted angle = 90°), refractive index of water is 1.33 and refractive index of air is 1.

(iv) Who observed induced current first time?

(v) Define electromagnet. How many poles it has?

(vi) What is the basic difference between generator and motor?

(vii) Write two properties of α-particles.

(viii) Write the difference between fission and fusion reaction.

Physics (Subjective) (PART - II) - Group - II

Time Allowed: 1.45 hrs. | **Maximum Marks:** 48

> Note: Attempt any TWO questions.

Q5(a). (Marks 4) How does electrostatic air cleaner work, explain.

Q5(b). (Marks 5) At one end of ripple tank 80 cm across, a 5 Hz vibrator produces waves whose wavelength is 40 mm. Find time the waves need to cross the tank.

Q6(a). (Marks 4) Define ultrasound and write any three its uses.

Q6(b). (Marks 5) The resistance of a conductor wire is 10MΩ. If a potential difference of 100 V is applied across its ends then find the value of current passing through it in mA.

Q7(a). (Marks 4) What is meant by nuclear transmutation? Give examples of Alpha, Beta and Gamma decay.

Q7(b). (Marks 5) An object 10.0 cm in front of a convex mirror forms an image 5.0 cm behind the mirror. What is the focal length of the mirror?